That's a good instinct. For a **first Minecraft mod**, building a full graph-based energy simulation is likely to turn into months of work before you ever get to the fun gameplay part.

I'd recommend simplifying the idea while keeping the interesting mechanic.

Option 1: Energy "pipes" with distance loss

The simplest implementation:

- Energy travels through cable blocks.
- Every cable segment causes a small loss.
- Machines only connect to adjacent cables.
- No network optimization, no routing.

Example:

```
Generator -- 10 cables -- Battery

1000 energy sent
1% loss per cable

1000 × 0.99^10 = 904 energy received
```

This can be implemented with a simple breadth-first search when energy is transferred.

Option 2: Throughput tiers instead of true efficiency curves

Instead of a complicated formula:

Transfer Rate	Loss per block
0-100/t	5%
100-1000/t	2%
1000+/t	0.5%

This creates the gameplay you want:

- Small trickles are inefficient.
- Large transmission lines are efficient.
- Easy to understand.
- Easy to code.

You don't need any fancy math.

Option 3: Voltage system (probably my favorite)

Treat throughput as "voltage."

- Low Voltage Cable
  - Cheap
  - High losses
- Medium Voltage Cable
  - Better efficiency
- High Voltage Cable
  - Very efficient
  - Expensive

Players naturally build transmission infrastructure without you needing to simulate real electrical engineering.

Many successful tech mods use this approach because it feels realistic while remaining simple.

What I'd avoid for a first mod

I would not start with:

- Dynamic network graphs
- Dijkstra routing
- Load balancing
- Multi-path flow simulation
- Chunk-spanning network managers
- Wireless transmission

Those are all things that sound cool but add a huge amount of debugging.

A first-mod roadmap

1. Add an Energy Cable block.
2. Add a Battery block.
3. Add a Creative Generator block.
4. Make energy move through connected cables.
5. Apply distance loss.
6. Add cable tiers.

At that point you'll already have a playable energy system and you'll learn most of the Fabric concepts:

- Blocks
- Block entities
- Networking/sync
- Saving data
- Ticking

That's a much more achievable first project than building something like a custom version of Flux Networks from scratch.

don't i get all of this for free with something like tech reborn?

Sort of. You can get a lot for free, but not necessarily the gameplay you're describing.

Tech Reborn already gives you:

- An energy API you can use instead of inventing your own. GitHub
- Energy storage, generators, machines, and cables.
- Energy transfer infrastructure.
- Integration with many Fabric tech mods via the Team Reborn energy ecosystem. GitHub +1

So if your goal is simply:

> "I want cables that move energy between machines."

Then yes, building a whole custom energy system is probably unnecessary.

The more important question is:

> What is the actual gameplay mechanic I want players to experience?

For example:

If you want distance loss

That already existed in older EU-based mods like IndustrialCraft 2, where cables lost energy over distance and different cable tiers had different efficiencies. Technic Pack +1

You could potentially build a small addon on top of Tech Reborn that modifies transfer efficiency rather than replacing the entire energy system.

If you want "high throughput is more efficient"

That's much less common.

You could:

- Use Tech Reborn's energy API.
- Create your own cable blocks.
- Implement your custom loss formula in those cables.

Then you're only writing:

- a few blocks,
- some transfer logic,
- and your balancing rules.

You aren't writing an entire energy ecosystem from scratch.

For a first mod, I'd seriously consider:

> **Use the Team Reborn Energy API, but make your own cables.**

That's a much smaller project than:

> **Create a brand-new energy type, storage system, transfer network, machine API, and compatibility layer.**

The first is a few thousand lines of code at most. The second can easily turn into a full tech mod that takes months.

What kind of mod are you envisioning? Is the energy system the whole point of the mod, or is it just infrastructure for machines you want to add later? That answer changes what I'd recommend.

Sources

The machines are honestly just for quality of life. The bulk of the mod is about managing and defending energy infrastructure.

That sounds much more focused, and honestly a lot more achievable than a full energy simulation.

The thing I like most is the **minimum throughput requirement**. That's a simple mechanic that naturally creates infrastructure planning.

For example:

Cable	Min Throughput	Loss
Copper	0 EU/t	5% / 100 blocks
Iron	100 EU/t	2% / 100 blocks
Gold	1000 EU/t	0.5% / 100 blocks

Now players have a real choice:

- A remote outpost producing 50 EU/t can't justify an iron line.
- A huge solar field producing 5000 EU/t absolutely should use gold.
- Mixing cable types becomes interesting.

That's easier to understand than some nonlinear efficiency formula while producing similar gameplay.

One thing I'd change

Instead of:

> energy doesn't flow at all below the minimum

I'd consider:

> energy below the minimum is heavily penalized

For example:

Gold cable

```
Target throughput = 1000 EU/t
```

At 100 EU/t:

- 90% loss

At 500 EU/t:

- 40% loss

At 1000 EU/t:

- normal efficiency

At 5000 EU/t:

- excellent efficiency

The reason is that hard cutoffs can feel confusing:

> "Why is my line not working?"

Whereas poor efficiency gives feedback:

> "It's working badly because I'm underutilizing it."

Batteries as infrastructure hubs

I think you're right that storage should probably be lossless.

The strategic choices are already:

- Where to place storage
- How much storage to build
- How much reserve capacity to keep

Adding storage loss usually just creates annoyance.

The battery monitor idea is strong

What you're describing sounds less like a battery and more like a **substation**.

A substation could:

Store energy

```
Current:
125,000 / 500,000
```

Show line statistics

```
North Line
+320 EU/t

Solar Line
+1200 EU/t

Factory Line
-900 EU/t

Mining Outpost
-150 EU/t
```

Graph history

```
Input
Output
Stored
```

Over:

- 1 minute
- 10 minutes
- 1 hour

Multiblock storage is probably a good fit

A single battery block is easy.

A battery bank is more interesting.

Something like:

```
Battery Controller

+ Battery Cells
+ Battery Cells
+ Battery Cells
```

Storage scales with:

- number of cells
- maybe different cell materials

This gives players infrastructure projects to build and defend.

The coolest part

The routing controls.

Imagine opening a substation and seeing:

```
Mining Base     10%
Factory         60%
City            30%
```

Or:

```
Factory
Max Draw: 2000 EU/t

Mining Base
Max Draw: 500 EU/t
```

Now when generation drops:

- critical systems stay online
- less important systems brown out

That creates meaningful management decisions without requiring complicated simulation.

For a first playable version

I'd honestly ship:

Cables

- Copper
- Iron
- Gold

Storage

- Battery Controller
- Battery Cells

UI

- Current stored energy
- Per-line throughput

And stop there initially.

The historical graphs and routing priorities can come later.

The danger with infrastructure mods is that it's easy to spend weeks building a beautiful monitoring UI before you've verified that "build transmission lines and defend them" is actually fun. The cable tiers and substations are enough to test the core gameplay loop.

What we have discussed so far isn't something that has already been built?

Not really, at least not as a complete package in the Minecraft mod ecosystem.

Pieces of it exist, but the combination you're describing is unusual.

Existing mods tend to fall into one of three camps

1. "Power is just a resource"
Examples include many RF/FE-based tech mods.

The network is mostly invisible:

- Connect cables.
- Energy arrives.
- Scale up generation.

The interesting part is the machines, not the grid.

2. "Power engineering"

Examples include older mods like IndustrialCraft 2.

These had:

- Cable losses
- Voltage tiers
- Transformers
- Exploding machines

But they generally didn't have:

- Throughput-driven efficiency economics
- Network monitoring as a central mechanic
- Infrastructure defense gameplay

The challenge was usually "don't blow up your machines."

3. "Logistics networks"
Examples include Applied Energistics 2 and similar systems.

These make the network itself interesting:

- Channels
- Routing
- Monitoring
- Network design

But they're about items and data, not power transmission.

What feels different about your idea

The thing that jumps out is that you're treating power infrastructure almost like a city-builder or utility-management game.

The core questions become:

- Where should power be generated?
- How should it be transmitted?
- What lines deserve expensive high-capacity infrastructure?
- How much reserve storage should exist?
- Which consumers get priority during shortages?
- What happens when part of the grid is destroyed?

That's much closer to games like:

- Factorio (logistics mindset)
- Workers & Resources: Soviet Republic

- Captain of Industry

than to most Minecraft tech mods.

The biggest thing I haven't seen

The battery/substation concept.

A lot of mods have batteries.

Very few make batteries into **operational control points** where you can:

- Monitor incoming lines
- Monitor outgoing lines
- View historical data
- Set priorities or quotas
- Act as regional grid hubs

That's the point where the infrastructure itself starts becoming gameplay.

The risk

There is one risk I'd watch for.

If all the player does is:

1. Build generator
2. Build cable
3. Build battery
4. Done

then you've created infrastructure without decisions.

The mechanics become interesting when there are tradeoffs:

- Copper vs iron vs gold
- Local generation vs long-distance transmission
- Cheap vulnerable lines vs expensive efficient lines
- Storage placement
- Power prioritization

As long as those tradeoffs exist, you're moving into territory that isn't already well-covered by existing Minecraft power mods.

So my answer would be:

> **The individual mechanics exist elsewhere, but the combination of throughput-**

sensitive transmission, grid management, monitoring substations, and infrastructure defense is not something I'd recognize as a standard Minecraft mod archetype.

That's a good place to be, because it means you're building on familiar concepts without simply recreating an existing mod.

So far I like the cables, but the battery idea seems too complicated. What is the simplest way for me to introduce that substation feel, where I am making concrete decisions about how to redistribute the power that has been gathered, without turning it into a spreadsheet simulator.

I think you're asking exactly the right question.

The danger isn't that the battery is too complicated to code. The danger is that it becomes a giant dashboard where the player spends time staring at numbers instead of making interesting decisions.

A good rule of thumb is:

> Players should make decisions about power, not manage power.

Simplest possible "substation"

What if a substation didn't store power at all?

Instead, it was just a routing block.

```
Generators
     ↓
Substation
  ↙  ↓  ↘
Mine Factory Base
```

The UI could be as simple as:

```
Mine:    Priority 1
Factory: Priority 3
Base:    Priority 2
```

When power is scarce:

1. Factory gets power first.
2. Base gets power second.
3. Mine gets whatever remains.

That's a meaningful strategic choice.

Capacity allocation

An alternative:

```
Mine:    200 EU/t max
Factory: 800 EU/t max
Base:    300 EU/t max
```

Now the player is deciding:

> How much of my generation budget should each region receive?

This feels like infrastructure planning rather than spreadsheeting.

Regional substations

This is the version I think best matches your vision.

Imagine every cable network eventually passes through substations.

```
Solar Farm
    ↓
North Substation
    ↓
Factory District

Wind Farm
    ↓
South Substation
    ↓
Mining District
```

The player isn't routing every machine.

They're routing entire regions.

That keeps complexity under control.

Throughput naturally creates decisions

Your cable system already provides a lot of gameplay.

Suppose:

- Copper is good for 0–100 EU/t
- Iron is good for 100–1000 EU/t
- Gold is good for 1000+ EU/t

Now the player is already making decisions:

> Do I aggregate power into a single gold trunk line?

or

> Do I keep generation local and use cheap copper?

You may not need much additional management.

The "breaker" idea

A really simple substation mechanic:

Each outgoing connection can be:

- Enabled
- Disabled

That's it.

Imagine a power shortage.

The player opens a substation and sees:

```
✓ Factory
✓ Base
✗ Decorative Lighting
✗ Rail Network
```

This feels like operating a power grid.

It's also extremely easy to understand.

The "three slots" approach

One design pattern that works well in Minecraft is forcing simplification through physical limits.

A substation might have:

- 1 input side
- 3 output ports

No giant network map.

No list of 50 consumers.

Just:

```
Output A
Output B
Output C
```

Players build additional substations if they need more complexity.

Minecraft is usually more fun when complexity comes from blocks in the world rather than controls in a GUI.

What I'd build first

If I were making this mod, I'd try:

Cable system

- Copper
- Iron
- Gold
- Throughput-dependent efficiency

Substation block

- No storage
- One incoming network
- Up to three outgoing networks

Controls

- Priority number OR on/off breaker

That's enough for situations like:

> "The solar farm can't power everything at night. Which district stays online?"

That's a concrete, understandable decision that feels like infrastructure management rather than accounting.

If that turns out to be fun, *then* you can think about adding storage, historical graphs, load shedding, backup generators, and all the other utility-grid features later. The priority system alone already gives players a reason to care about how power is distributed.

Okay I have an idea. We add a monitor block. The monitor can be attached to any cable and indicates whether it is being underutilized, which direction power is flowing, and right clicking it gives a graph of the power flow through that cable over some fixed timeframe, 3 in-game days? Three tiers of batteries are added and they are lossy, so storing energy is something to avoid, but the battery allows you to set a target rate for one output cable.

I actually like the monitor block more than the substation idea.

The reason is that it keeps the decision-making **in the world** instead of in a GUI. Players walk their grid, inspect bottlenecks, and upgrade infrastructure. That's very Minecraft.

The monitor block

A monitor attached to a cable could show:

- Flow direction (arrow texture or particle effect)
- Current throughput
- Utilization state

Something like:

```
Gold Cable
Flow: 850 EU/t →
Utilization: 42%
Status: Underutilized
```

That's immediately understandable.

Underutilization is an excellent signal

The player doesn't need to know your exact formulas.

They only need:

- Red = overloaded
- Green = optimal
- Blue = underutilized

Then they start asking:

> Why am I paying for gold cable when I'm only sending 100 EU/t through it?

which is exactly the infrastructure-planning behavior you want.

Three-day graphs are probably enough

I wouldn't make them configurable initially.

Just:

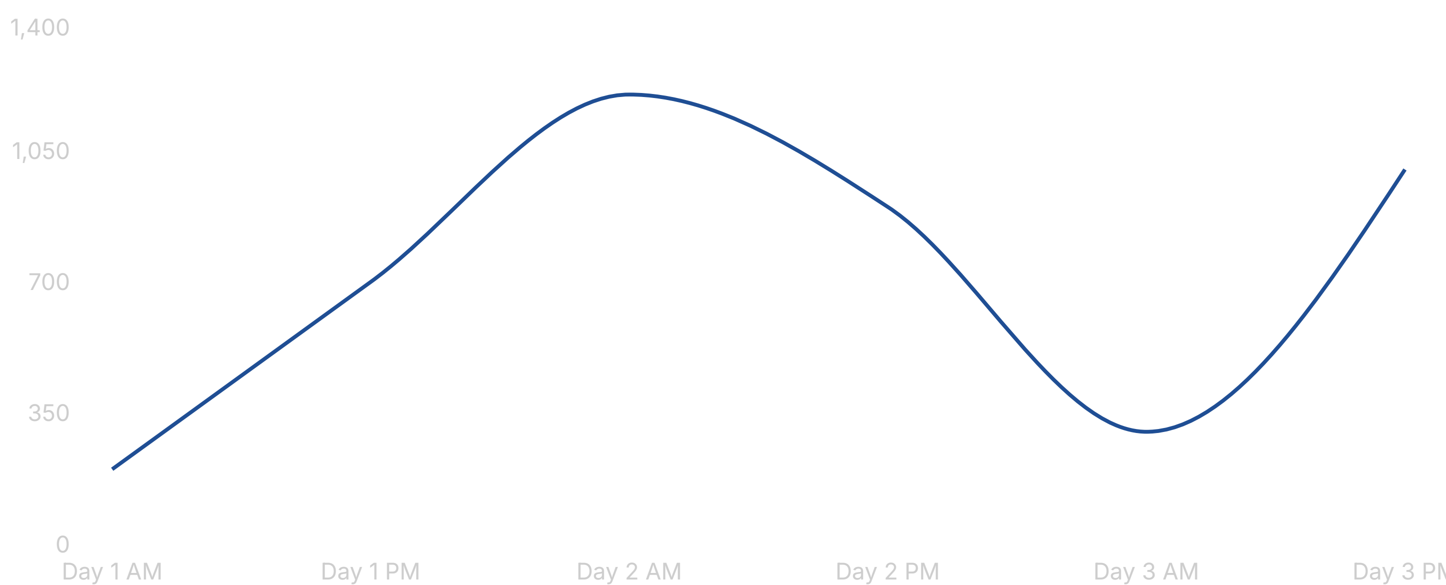
```
Last 3 Minecraft days
```

and a line graph.

For example, a monitor UI might show something like:

Cable throughput

Flow over the last three in-game days.



You obviously wouldn't hardcode the data; this is just the kind of information I'd expect the player to see.

The graph gives them evidence that:

- a solar farm spikes during the day,
- a mine runs continuously,
- a line is oversized or overloaded.

About lossy batteries

I was initially skeptical, but with your updated design I think they make more sense.

The key is that the battery isn't just storage.

It's a **power conditioner**.

Suppose:

```
Wind Farm
      ↓
Battery
      ↓
Gold Line
```

The battery allows:

```
Target Output: 1000 EU/t
```

Now the battery:

- buffers fluctuations,
- guarantees throughput,
- keeps the gold line operating efficiently.

The loss is the price paid for stabilization.

That's a meaningful tradeoff.

I'd make the loss happen only when cycling energy

For example:

- Stored energy doesn't decay.
- Charge/discharge incurs losses.

```
Input 1000
Stored 950
Output 900
```

or

```
95% charge efficiency
95% discharge efficiency
```

This feels intuitive.

Players won't lose power because they forgot about a battery for three days.

They lose power because they're actively using storage as infrastructure.

Battery tiers

These could differ by three parameters:

Tier	Capacity	Efficiency	Max Output
Copper	Small	80%	200 EU/t
Iron	Medium	90%	1000 EU/t
Gold	Large	97%	5000 EU/t

Notice that now the tiers naturally complement the cable tiers.

The target-output mechanic is the interesting part

This is where the "substation" feeling emerges.

Imagine:

```
Battery A
Target Output: 2000 EU/t
```

The battery will:

- discharge when generation falls below 2000,
- charge when generation exceeds 2000.

The player is effectively saying:

> "This district should receive 2000 EU/t."

That's a strategic decision without requiring a giant routing interface.

What I think the gameplay loop is becoming

1. Build generators.
2. Aggregate generation.
3. Use monitors to find inefficient lines.
4. Upgrade transmission infrastructure.
5. Use batteries to smooth generation and maintain target throughput.
6. Defend important lines and storage.

That's a pretty coherent loop, and importantly, each mechanic teaches the player something:

- Monitors answer "What is happening?"
- Cables answer "How should I transmit power?"
- Batteries answer "How stable should this district be?"

For a first mod, that's starting to feel like a complete design rather than just a collection of blocks. The monitor especially feels like a feature that players would immediately understand and use.

I think that the additions are quite minimal as well, fundamentally only 3 new blocks aer being added, cables, monitors and batteries, including the tiers, 7 and the resulting gameplay feels quite rich.

I agree, and that's usually a good sign.

A lot of first-time modders think richness comes from adding dozens of blocks. In practice, some of the most memorable Minecraft mechanics come from a very small number of blocks interacting in interesting ways.

Looking at your design:

Blocks

Cable

- Copper
- Iron
- Gold

Battery

- Copper
- Iron
- Gold

Monitor

- One block

That's 7 blocks total.

But the decisions multiply

A player building a power line now has to think about:

- Do I use copper, iron, or gold?
- Am I generating enough power to justify the higher tier?
- Should I centralize generation or keep it local?
- Is this line underutilized?
- Is this line overloaded?
- Should I buffer this network with a battery?
- How much output should the battery target?

Those decisions emerge from interactions between the blocks rather than from having lots of block types.

The monitor is doing a lot of work

One thing I particularly like is that the monitor doesn't *change* the network.

It reveals information.

That's often the hallmark of a strong management mechanic.

Without monitors:

```
Power line works
Power line doesn't work
```

With monitors:

```
Power line is overloaded
Power line is underutilized
Power line has fluctuating demand
Power line only spikes during the day
```

Now players can diagnose problems instead of guessing.

The battery is pulling double duty

What you've accidentally stumbled into is that the battery serves two roles:

1. Storage
2. Flow regulation

If it can maintain a target output rate, it effectively becomes your "substation" without needing a separate substation block.

That's a huge simplification.

I would strongly resist adding a dedicated substation now.

Every time you think:

> "Maybe I need a substation block..."

ask:

> "Could the battery already do that?"

If the answer is yes, keep the battery.

The strongest part of the design

I think the strongest part is actually this combination:

- High-tier cables want high throughput.
- Batteries can stabilize throughput.
- Batteries incur losses.

That creates a natural tradeoff:

```
Direct transmission
↓
Efficient
but unstable

Battery-buffered transmission
↓
Stable
but lossy
```

That's a real decision.

And it's understandable without reading a wiki.

One thing I'd be careful about

Don't let the monitor turn into an industrial control panel.

For example, I'd avoid:

- 20 statistics
- Per-tick data
- Complex routing tables
- Configurable graph ranges
- Forecasting

At least initially.

A monitor that shows:

```
Current Flow
Average Flow
Utilization
3-Day Graph
```

is probably enough.

Players should be able to understand a line's health in a few seconds.

A good litmus test

Imagine explaining the entire mod in one sentence:

> "Build efficient power grids by matching cable tiers to throughput, using batteries to stabilize supply, and monitors to identify bottlenecks."

That's surprisingly concise.

If you can explain the core gameplay that quickly and still imagine players making meaningful decisions for hours, you're probably onto a solid design. Many successful mods have started from less.